

Coding & Solution

Date	8 JULY 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Bhagya1416/Rising-Waters
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	• Historical weather data preprocessing. • Flood prediction using Decision Tree, Random Forest, KNN, and XGBoost. • Selection of the best-performing model (XGBoost – 96.55% accuracy). • Flask-based web application for user input and prediction. • User-friendly interface for disaster management authorities. • Support for future deployment on IBM Cloud.
Pending / Incomplete Features	• Integration with live weather APIs. • SMS/Email alert notifications. • Interactive GIS map visualization. • Mobile application support.
Setup / Run Instructions	1. Install Python and required libraries (Flask, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib). 2. Clone the GitHub repository. 3. Install dependencies using <code>pip install -r requirements.txt</code> (if available). 4. Run the Flask application using <code>python app.py</code>. 5. Open the provided local URL (e.g., <code>http://127.0.0.1:5000</code>) in a web browser and enter weather details to obtain flood predictions.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: ☐ **The project follows clean coding practices with a modular structure.**

- ☐ Multiple Machine Learning algorithms (Decision Tree, Random Forest, KNN, and XGBoost) were evaluated, with XGBoost achieving 96.55% accuracy.
- ☐ The Flask web application provides a simple and responsive interface for flood prediction.
- ☐ The solution is designed to support early flood warning, resource allocation, and disaster preparedness.
- ☐ The architecture allows future enhancements such as real-time weather API integration, cloud deployment, and automated alert systems.